A Moon City for a Million People? Water Might Run Out

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Could one million people live on the Moon? Two scientists checked the numbers. Their answer is yes, but only for a short time. Power is not the problem. Water is.

SpaceX chief Elon Musk has said his company wants to build a city on the Moon that grows by itself in less than 10 years. A Mars city would take more than 20 years, he said. Blue Origin chief Jeff Bezos has said millions of people will live in space by 2045. Martin Elvis, an **astrophysicist** at Harvard University's Smithsonian Astrophysical Observatory, and Jonathan McDowell, an **astrophysicist** at Durham University in England, wanted to test those ideas.

In a paper in the journal Frontiers in Space Technologies, they found that solar and **nuclear** power could support one million people on the Moon. But there is not enough water. If the Moon holds about one billion **tonnes** of water, a city of one million people could last only about 100 years, and only if 98 percent of the water is **recycled**, the same rate as on the International Space Station. A town of 10,000 people could last for **centuries**. But Elvis said the one-billion-tonne guess is probably far too high, perhaps 20 or 30 times too big. So do not start packing just yet.

Word Treasure

astrophysicist -- A scientist who studies stars, planets, and space.
nuclear -- A kind of power made by splitting tiny parts inside atoms.
recycled -- Used again instead of being thrown away.
tonnes -- A unit for measuring very heavy weights.
centuries -- Periods of one hundred years.

Background you need

A city on the Moon would need power, air, food, and water for everyone who lives there.

Power can come from the Sun and from nuclear power, but water is much harder to find on the Moon.

People could reuse almost all their water, the way crews do on the International Space Station, yet even that may not be enough for a very large city.

Break it down

WHO: Martin Elvis and Jonathan McDowell, two scientists, and company chiefs Elon Musk and Jeff Bezos.

WHAT: They tested whether a city of one million people could live on the Moon and found that water would run out.

WHERE: The Moon.

WHY: Because there is not enough water on the Moon, even when almost all of it is recycled.

Why it matters

It shows why dreaming big still needs careful checking, just like making sure you have enough water before a long trip.

Test what you remember

1. What did the two scientists check in their study?

- (A) Whether the Moon has its own weather
- (B) How fast a rocket can reach the Moon
- (C) How many stars are near Earth
- (D) Whether one million people could live on the Moon

2. According to the scientists, what is the main problem for a Moon city?

- (A) Food
- (B) Water
- (C) Power
- (D) Air

3. What kinds of power could support one million people on the Moon?

- (A) Power from lightning
- (B) Solar and nuclear power
- (C) Coal and oil power
- (D) Wind and water power

4. If the Moon holds about one billion tonnes of water and almost all of it is recycled, how long could a city of one million people last?

- (A) About ten years
- (B) About 100 years
- (C) About fifty years
- (D) For thousands of years

5. How long could a town of 10,000 people on the Moon last?

- (A) Less than a year
- (B) Only one winter
- (C) Only a few weeks
- (D) For centuries

6. Who said his company wants to build a city on the Moon that grows by itself?

- (A) Jonathan McDowell
- (B) Martin Elvis
- (C) Jeff Bezos
- (D) Elon Musk

Answer key (try the quiz first!): 1: D 2: B 3: B 4: B 5: D 6: D

Questions to think about

1. Elon Musk: He wants his company to build a city on the Moon that grows by itself in less than 10 years.
2. Jeff Bezos: He has said millions of people will live in space by 2045.
3. Martin Elvis: He said the one-billion-tonne guess for the Moon's water is probably far too high.

MY THOUGHTS (at least 20 words):